INCH-POUND

MIL-S-13949/28A
18 May 1993
SUPERSEDING
MIL-S-13949/28
4 October 1991

MILITARY SPECIFICATION SHEET

SHEET, PRINTED WIRING BOARD, PREPREG, BASE MATERIAL SC
(WOVEN S-2 GLASS FIBER 1/ REINFORCEMENT, MAJORITY CYANATE
ESTER RESIN, FLAME RESISTANT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-13949.

REQUIREMENTS:

Reinforcement material: S for woven electrical grade S-2 glass fiber. 1/ This reinforcement material is a continuous filament yarn with a chemical composition by weight that is within the following limits:

Al_2O_3	24 - 26 percent
SiO_2	64 - 66 percent
MgO	9 - 11 percent
Na_2O and K_2O	0 - 0.3 percent
Fe_2O_3	0 - 0.3 percent

Composition to be certified by the yarn supplier as requested. All other parameters shall be as detailed in table I.

Weave style:

Plain weave: A fabric configuration where each warp end should go over one pick and under the next, and each pick should go over one warp end and under the next.

Satin weave: A fabric configuration where the surface is almost made up entirely of warp or filling floats over all but one of a designated number of adjacent yarns, thereby producing a smooth surface. The intersection points do not fall in a straight diagonal (or twill), but in pattered formation. For example, in a four harness satin weave pattern, there is a three by one interlacing, whereby, a filling yarn floats over three warp yarns and under one.

Resin system: C in accordance with MIL-S-13949.

1/ The term "S-2 glass fiber" is a registered trademark of the Owens-Corning Fiberglass Corporation. The term "S-2 glass fiber" is used herein in a generic sense and not in the sense of a registered trademark to identify a type of reinforcement material used in the production of printed wiring board materials. The use of the term "S-2 glass fiber" herein does not impose a requirement to utilize Owens-Corning's "S-2 glass fiber" material in the production of printed wiring board materials manufactured in accordance with this specification sheet, although the use of the Owens-Corning material would meet this specification sheet's requirement for an "S-2 glass fiber" reinforcement material.

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TABLE I. Woven reinforcement construction characteristics.

Style	Thickness (inches)	Thread count	Weight, maximum (±10 percent) (ounces per square yard)
6080	.0021	60 x 47	1.38
6112	.0036	40 x 39	2.08
6116	.0040	60 x 58	3.09
6120 1/	.0038	60 x 58	3.09
6180 1/	.0020	60 x 60	1.55
6313	.0035	60 x 64	2.38
6314 1/	.0033	60 x 64	2.38
6413	.0034	60 x 56	2.30
6628	.0068	44 x 32	6.03

1/ These styles are a four harness satin weave. All other styles are plain weave.

Special consideration (third letter designator): N in accordance with MIL-S-13949.

Qualification inspection routine. The inspection routine shall include the glass transition temperature test in accordance with MIL-S-13949.

Conditioning of specimens prior to test: Unless otherwise specified herein, specimens shall be conditioned in accordance with the applicable test method.

Property values:

Flammability, duration of burning time, each flame application: 15 seconds maximum.

Volatile content (conditioning of specimen prior to test, E-0.25/145): 2.0 percent.

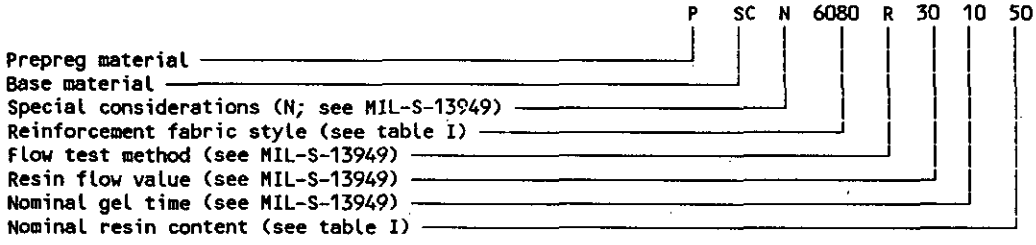
Electrical strength, average: 750 volts/mil minimum.

Loss tangent at 1 MHz: .015 maximum.

Permittivity at 1 MHz: 4.3 maximum.

Supersession information: Not applicable.

Type designation: Consists of "PSC" prefix followed by coded alphas and numeric characters as shown in the following example:



Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 99
NSA - NS
DLA - ES

User activities:
Army - AT, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5998-0032-03)

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